

# A Multilingual Transformer Framework for Sentiment & Emotion Detection in E-commerce with Conceptual Bot Deployment

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**Abstract**— The rapid growth of e-commerce platforms in multilingual regions such as Bangladesh has resulted in large volumes of customer reviews and comments expressed in different mixed text. However, existing sentiment analysis and emotion detection systems often perform poorly in these contexts due to limited coverage for low-resource languages. While deep learning and transformer models have shown promise in multilingual NLP, their application to diverse linguistic settings remains underexplored. To bridge this gap, we construct a comparative framework for multilingual NLP pipelines targeting binary sentiment analysis and multi-class emotion detection in e-commerce reviews. A total of 63,942 reviews were collected, preprocessed, and divided into four language-based subsets to evaluate model's robustness under varying linguistic conditions. Six models SVM, Bi-LSTM, mBERT, XLM-RoBERTa, BanglishBERT, and DistilBERT were trained and tested across these subsets, with performance assessed using accuracy, precision, recall, and weighted F1-score. Results show that transformer models consistently outperformed traditional approaches, with XLM-RoBERTa achieving the best results, 92.6% for sentiment analysis, 68.9% for emotion detection. Based on these findings, we propose a smart e-commerce bot architecture that integrates multilingual sentiment and emotion extraction to enable real time, scalable customer service, automatic assistance and product evaluation for business improvement.

**Keywords**— *Multilingual NLP, Sentiment Analysis, Emotion Detection, Transformer Models, E-commerce Bot.*

## I. INTRODUCTION

As online Shopping and marketing are growing rapidly, customer reviews increasingly become an information source for both consumers and merchants. Automated sentiment analysis enables opinion classification into positive or negative labels, and emotion detection provides an even more fine-grained view through the identification of feelings such as happiness, sorrow, anger. Both tasks together enable firms not only to quantify customer satisfaction but also derive emotional states that inform product improvement, personalized services and targeted marketing campaigns. The most recent papers highlight the fact that the combination of

sentiment and emotion analysis is becoming increasingly important for the understanding of the behavior of consumers within online commerce [1],[2].

Although there is significant worldwide development of sentiment analysis, most studies considered the English corpora alone, and Bangla and Bangla-English (Banglish) reviews remain under researched despite their very high prevalence within South Asian e-commerce markets[3]. Both languages carry with them the complexities of transliteration, informal writing, and extensive code-switching that can lower the performance of standard models.

Some studies attempted to fill such gaps. As for the monolingual Bangla context, Kabir et al. (2023) introduced the large-scale dataset of BanglaBook and claimed that transformer-based models such as BanglaBERT outperform traditional models like SVM and LSTM[1]. Similarly, Hasan et al. (2023) found that transfer learning and transformer-based models significantly improve sentiment classification for Bangladeshi e-commerce reviews[4]. Following up on that, Alam et al. (2024) introduced more than 203K comments dataset for Bangla, BangDSA, and developed skipBanglaBERT with CNN & Bi-LSTM models and claimed state of the art document level sentiment accuracy of 90.24% [5].

In code-mixed cases, Tareq et al. (2023) used data augmentation with XGBoost and FastText embeddings for code-mixed Bangla-English reviews and observed significant performance improvements achieving a weighted F1 score of 87% [6]. Likewise, the SentMix-3L dataset presented code-mixed Bangla-English-Hindi reviews and set up transformer-based baselines[7]. and the Semantics Squad team presented BanglaBERT Small reaching 71.33% accuracy[8]. Emotion classification for Bangla remains limited. Ahmed et al. (2024) analyzed social media emotion dynamics of Bangla via Random Forest and reported an accuracy of 84% [9]. Similarly, Sadaf et al. (2023) created an affect database and gave benchmarks for multi-class emotion classification, The findings demonstrate that Logistic Regression, SVM, and Naïve Bayes yielded the best performance, with Logistic Regression (BOW + Unigram) achieving an accuracy of 68.8% and a macro F1-score of 65%.[10]. In addition to e-

commerce, some comparative analysis of multilingual approaches for classifying both the sentiment and the text of an examined multilingual corpus, where Manias et al. (2023) employed transformer models for the identification of multilingual sentiment, achieves 75.85% accuracy with 76.42% F1 score [11].

From the foregoing studies, there are two significant gaps identified: (1) few studies on code-mixed Banglish reviews for e-commerce, and (2) few studies on sentiment–emotion detection systems that could be deployed over multilingual user-generated text. Closing such gaps is vital for online platforms like Daraz, Pickaboo, and Evaly for which reviews are always written in Bangla or Banglish. That motivates our sentiment and emotion detection work under the multilingual scenario with the most recent transformer architecture.

This paper presents a comprehensive framework for sentiment & emotion detection for multilingual e-commerce reviews, using both traditional machine learning & transformer based deep learning models. The main goal of this effort is to use robust sentiment & emotion recognition to enhance customer understanding in online marketplaces. The key contributions of this paper are as follows:

1. A multilingual sentiment and emotion classification framework is developed and trained on a large dataset comprising reviews written in Bangla, English, Banglish and code-mixed in order to better understand customer thoughts across linguistic boundaries.
2. Dual level analysis of customer feedback is made possible by the dataset's diversity, which includes both sentiment (positive and negative) and emotion (happy, love, sadness, rage, and fear) annotations.
3. A novel comparative study of SVM, Bi-LSTM, mBERT, BanglishBERT, DistilBERT and XLM-RoBERTa was conducted across four language subsets, demonstrating XLM-RoBERTa's superior performance in cross-domain sentiment & emotion detection.
4. Design & proposal of a multilingual emotion aware bot for e-commerce platforms that combines intelligent response generation with personalized recommendations and product level insights from both the customer & seller's perspective.

## II. METHODOLOGY

This section describes full implementation of the proposed e-commerce assistant system ranging from data collection to preprocessing, model training and evaluation using different models to find our best model for implementation.

### A. Dataset Collection

The dataset was collected from Rashid et al. (2024)[12]. The dataset consists of 78,130 product reviews collected from two Bangladeshi e-commerce platforms, Daraz and Pickaboo. All reviews are annotated with two types of classification. Sentiment analysis part has binary classification “Positive” & “Negative” and classifying into 5 classes: ‘Happy’, ‘Sadness’, ‘Love’, ‘Anger’, ‘Fear’. After preprocessing & deleting duplicates, our final dataset included 63,942 samples where majority of portion was positive reviews with 53,525 positive (83.70%) and 10,417 negative reviews (16.30%). To address emotion distribution, the class “Happy” had the major portion with 37,548 samples, the other classes: Love (15,949), Sadness (6,281), Anger (3,042), and Fear (1,122).

TABLE I. DATASET DISTRIBUTION BASED ON CLASSES

| Analysis  | Class    | Numbers | Percentage |
|-----------|----------|---------|------------|
| Sentiment | Positive | 53525   | 83.70%     |
|           | Negative | 10417   | 16.30%     |
| Emotion   | Sadness  | 6281    | 9.82%      |
|           | Anger    | 3042    | 4.76%      |
|           | Fear     | 1122    | 1.75%      |
|           | Happy    | 37548   | 58.72%     |
|           | Love     | 15949   | 24.95%     |

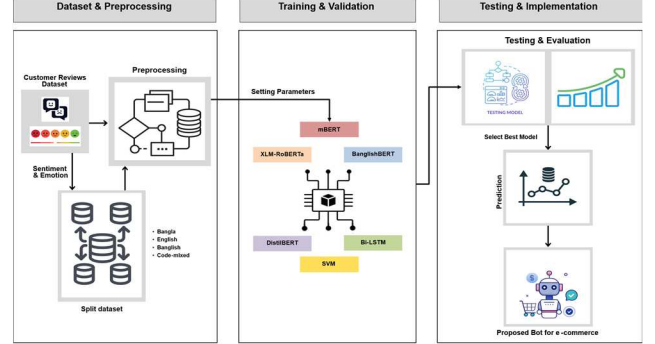


Fig. 1. Methodology of workflow

Fig. 1 shows the methodology of our workflow. Collecting dataset & preprocessing part was done considering different techniques for linguistic diversities in the dataset. After setting all the parameters, we trained & validated with six different machine learning & transformer-based models. Then we evaluated all the model's performance to find the best model for implementation & further analysis.

### B. Preprocessing

Fig. 2 represents the structured preprocessing pipeline applied to the dataset to make it more generalized for training & evaluating the necessary measurement parameters.

We removed the duplicate reviews from the dataset. Then we cleared the noises such as emoji though sometimes it can be meaningful but the large dataset doesn't contain enough emoji's and didn't affect that much in the ultimate results. We also removed unnecessary punctuations, URLs & HTML tags, removed stopwords considering the negations as it is necessary to determine the sentiment. In addition to standard cleaning steps, we incorporated a discrepancy removal phase to eliminate mismatches between review and annotated labels & ensures consistency. Tokenization was done considering different languages. Verb specific stemming & POS based lemmatization were done for English languages. Then cleaned tokens were rejoined and saved for model input.

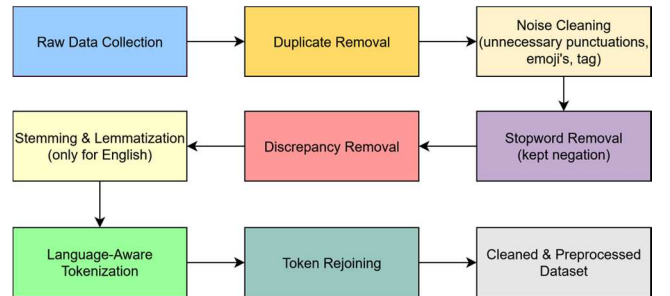


Fig. 2. Data Preprocessing pipeline.

### C. Model Training

We implemented six models to evaluate our multilingual sentiment & emotion classification performance over our final dataset. The dataset was split into train, test, validation set.

80% of the entire dataset was taken for train set, 10% for each testing & validation. We used SVM & Bi-LSTM as a benchmark model against the powerful transformer-based models. Four transformer-based BERT models were implemented to evaluate & compare the performance across different languages & tasks to find the best model for our approach. Learning rates ranging from  $1e^{-5}$  to  $5e^{-5}$  and batch sizes of 16, 32, and 64 were evaluated. After comparative analysis, the best results were achieved using a learning rate of  $2e^{-5}$  and a batch size of 32. The AdamW optimizer was employed for training. Cross-Entropy Loss was used to measure the discrepancy between the model's predicted output and the true labels. Based on validation loss increase with training loss we applied early stopping to find the best epoch for our design & eliminate the risk of overfitting. Among all models, XLM-RoBERTa has achieved the highest accuracy & robust generalization for both the sentiment & emotion detection.

Fig. 3 shows the new XLM-RoBERTa model architecture, comprising three parts: embedding layer, Transformer encoder, and classification heads. The embedding layer includes token and positional embeddings. The Transformer encoder developed contextual features using stacked multi-head self-attention blocks (12-24) with feed-forward networks & normalizations. Classification heads predict sentiment and emotion labels.

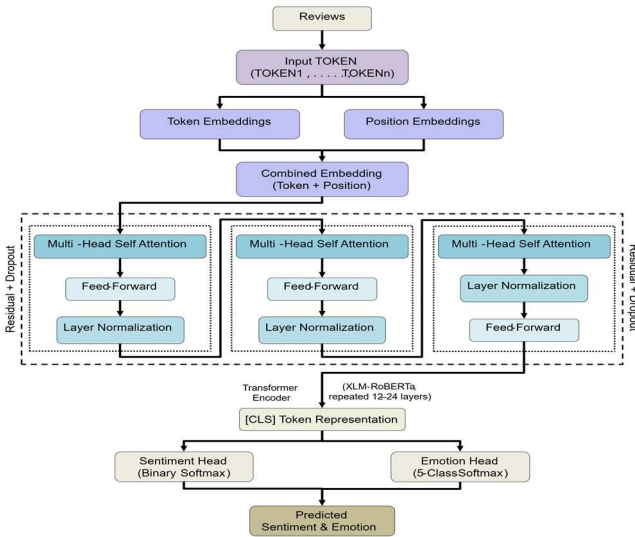


Fig. 3. Architecture of XLM-RoBERTa model for ecommerce system.

#### D. Model Evaluation

Standard classification metrics were used to evaluate the trained models: Accuracy, Precision, Recall, F1-score, and class-wise support across both sentiment detection for binary classification and emotion detection for multi-class classification tasks. The best model was selected based on validation set performance by carefully monitoring the validation loss to prevent overfitting. Confusion matrices were also evaluated thoroughly to identify and analyze misclassifications. Though mBERT and DistilBERT performed reasonably well. Overall, XLM-RoBERTa was slightly better in terms of validation and testing accuracy, losses, and consistently balanced predictions for both the sentiment and emotion classes across the main dataset and multilingual subsets of the dataset, which ultimately made it the most suitable choice for deployment and further detailed analysis.

### III. MODEL RESULT & DISCUSSION

#### A. Results

In our study, Accuracy, precision, recall, F1-score, were measured and compared for six different NLP related models: SVM, Bi-LSTM, mBERT, XLM-RoBERTa, BanglishBERT and DistilBERT. These metrics will show the basis of how each model performs in terms of sentiment & emotion detection regarding both accuracy and F1 score, two very important factors in real time text classification systems.

TABLE II. PERFORMANCE EVALUATION OF DIFFERENT MODELS FOR SENTIMENT ANALYSIS

| Model        | Accuracy    | Weighted F1 | Precision   | Recall      |
|--------------|-------------|-------------|-------------|-------------|
| SVM          | 0.89        | 0.88        | 0.88        | 0.89        |
| Bi-LSTM      | 0.90        | 0.90        | 0.90        | 0.90        |
| mBERT        | 0.93        | 0.92        | 0.92        | 0.93        |
| XLM-RoBERTa  | <b>0.93</b> | <b>0.93</b> | <b>0.92</b> | <b>0.93</b> |
| BanglishBERT | 0.88        | 0.88        | 0.87        | 0.88        |
| DistilBERT   | 0.92        | 0.92        | 0.92        | 0.92        |

TABLE III. PERFORMANCE EVALUATION OF DIFFERENT MODELS FOR EMOTION DETECTION

| Model        | Accuracy    | Weighted F1 | Precision   | Recall      |
|--------------|-------------|-------------|-------------|-------------|
| SVM          | 0.64        | 0.58        | 0.61        | 0.64        |
| Bi-LSTM      | 0.65        | 0.61        | 0.66        | 0.63        |
| mBERT        | 0.68        | 0.67        | 0.67        | 0.68        |
| XLM-RoBERTa  | <b>0.69</b> | <b>0.67</b> | <b>0.67</b> | <b>0.69</b> |
| BanglishBERT | 0.63        | 0.57        | 0.61        | 0.63        |
| DistilBERT   | 0.67        | 0.66        | 0.65        | 0.67        |

Table-II & Table-III presents a comparison of accuracy, precision, recall and weighted F1 score of different models for our sentiment analysis & emotion detection task. Here in table-II, XLM-RoBERTa clearly surpasses the other models in terms of all the metrics with accuracy of 93%, precision 92%, recall 93% & most importantly with 1% higher weighted F1 score than mBERT makes it feasible for our project. DistilBERT performs almost similar & faster than the other models but after analyzing all the parameters XLM-RoBERTa performs better comparing to the other models.

In Table-III, we can see same type of performance parameters for emotion detection where XLM-RoBERTa gives 69% accuracy & recall which is 1% higher than the second best mBERT model despite weighted F1 & precision is found similar for both models. DistilBERT's performance was quite impressive in terms of speed in our training & testing period but not reached similar accuracy like XLM-RoBERTa model.

TABLE IV. TRAINING VS VALIDATION VS TESTING ACCURACY FOR SENTIMENT ANALYSIS & EMOTION DETECTION

| Model        | Sentiment         |                     |                  | Emotion           |                     |                  |
|--------------|-------------------|---------------------|------------------|-------------------|---------------------|------------------|
|              | Training Accuracy | Validation Accuracy | Testing Accuracy | Training Accuracy | Validation Accuracy | Testing Accuracy |
| SVM          | 0.915             | 0.889               | 0.889            | 0.685             | 0.642               | 0.636            |
| Bi-LSTM      | 0.909             | 0.899               | 0.904            | 0.653             | 0.652               | 0.656            |
| mBERT        | 0.934             | 0.926               | 0.925            | 0.678             | 0.676               | 0.685            |
| XLM-RoBERTa  | <b>0.940</b>      | <b>0.927</b>        | <b>0.926</b>     | <b>0.687</b>      | <b>0.688</b>        | <b>0.689</b>     |
| BanglishBERT | 0.892             | 0.891               | 0.883            | 0.628             | 0.622               | 0.629            |
| DistilBERT   | 0.933             | 0.924               | 0.920            | 0.681             | 0.684               | 0.669            |

Table-IV represents the training, validation and testing accuracy of our tasks across different models. In our both tasks, the results were quite similar for all the models we have trained our dataset. Among them, XLM-RoBERTa achieved

the highest accuracy for training (94%), validation (92.7%) & testing (92.6%) in sentiment analysis. mBERT got almost similar with just 0.1% less accuracy for validation & testing accuracy than XLM-RoBERTa. Again showing the accuracy for emotion task where the overall highest accuracy achieved by XLM-RoBERTa is 68.7% for training, 68.8% for validation, 68.9% for testing which is slightly outperformed mBERT (0.678/0.676/0.685) & DistilBERT (0.681/0.684/0.669). BanglishBERT showing the lowest performance among all the models while Bi-LSTM, SVM performed moderately good.

### B. Performance analysis of XLM-RoBERTa model

Next the precision, recall, F1 score and support, of the XLM-RoBERTa model in detecting classes of sentiment & emotion to further evaluate its performance.

Table-V shows how well sentiment & emotion classification works for different classes. Precision, Recall, F1 score & Support are used as metrics to evaluate the performance of XLM-RoBERTa model. As most of the reviews from the dataset were positive reviews so the model performs well for positive class with an impressive precision (95%), Recall (96%), F1 score (96%) & support 5356. There were low number of samples for negative reviews so overall precision (79%), recall (75%) & F1 score (77%) was slightly less than positive as we didn't give any weight to the classes which have less samples in our dataset. For Emotion detection, there has strict differences of the class-wise performance as it is an imbalanced dataset where the number of samples for "happy" is significantly higher than the other classes.

TABLE V. CLASS-WISE PERFORMANCE EVALUATION OF XLM-ROBERTA MODEL FOR SENTIMENT & EMOTION DETECTION

| Analysis  | Class    | Precision | Recall | F1 Score | Support |
|-----------|----------|-----------|--------|----------|---------|
| Sentiment | Positive | 0.95      | 0.96   | 0.96     | 5356    |
|           | Negative | 0.79      | 0.75   | 0.77     | 1044    |
| Emotion   | Happy    | 0.75      | 0.84   | 0.79     | 3840    |
|           | Love     | 0.60      | 0.44   | 0.51     | 1516    |
|           | Sadness  | 0.50      | 0.58   | 0.54     | 607     |
|           | Anger    | 0.57      | 0.44   | 0.49     | 333     |
|           | Fear     | 0.38      | 0.06   | 0.10     | 104     |

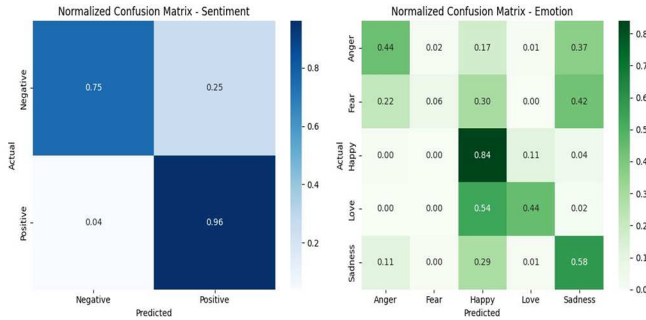


Fig. 4. Normalized confusion matrix for XLM-RoBERTa model.

The confusion matrices of our top-performing model, XLM-RoBERTa, on the dataset are shown in Fig. 4. The model's classification performance across several sentiment and emotion categories can be better understood by examining the confusion matrices. It highlights potential areas for improvement by providing a thorough analysis of the model's predictions in comparison to the actual labels. When it comes to accurately identifying instances of "Happy" and "Love," XLM-RoBERTa performs the best. Nonetheless, there are noteworthy cases where "Sadness" and "Fear" are confused, indicating that it may be difficult to tell these two feelings

apart.

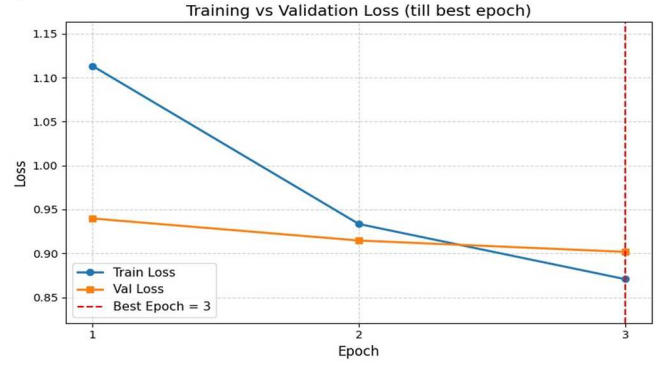


Fig. 5. Training vs validation loss curve for XLM-RoBERTa model.

Fig. 5 showing the training vs validation loss across epoch numbers to get the best epoch for our evaluation. Early stopping condition was applied to prevent overfitting issue.

### C. Model's performance on multilingual diversity

We split the dataset into 4 different language's subsets to evaluate the model's performance across linguistic diversity & found Bangla (9,605), English (16,500), Banglish (14,281), and Code-Mixed (23,556) samples.

From table-VI & VII, the accuracy & weighted F1 score of different models across different languages are shown for sentiment & emotion task. We observed the traditional machine learning model (SVM) & deep learning model (Bi-LSTM) gives moderate results. The most notable advancements are found in transformer-based models.

For code-mixed, Banglish & English languages, XLM-RoBERTa achieved both the highest accuracy (91%) & weighted F1(91%) over the other models but it fails to give better accuracy & F1 score for Bangla language as mBERT & BanglishBERT gives the best accuracy & BanglishBERT performs with highest F1-score in sentiment case. we found most accuracy with 96% & same percentage for weighted F1 score in English language with XLM-RoBERTa model. On the other hand, the best accuracy of emotion detection was found for code-mixed language of 68% with a F1 score of 67% by XLM-RoBERTa model just slightly better than other models. mBERT performs better in Bangla case with an accuracy of 67% & weighted F1 of 65%. BanglishBERT's performance is also noteworthy in Bangla case. Due to its pre-training on Bengali and English data rather recognizing Banglish data specifically, BanglishBERT did not perform well as expected on the Banglish dataset. Conversely, mBERT performs comparable to XLM-RoBERTa in both sentiment & emotion case.

TABLE VI. ACCURACY & WEIGHTED F1 OF DIFFERENT MODELS FOR MULTILINGUAL SENTIMENT ANALYSIS

| Model         | Accuracy     |             |             |             | Weighted F1  |             |             |             |
|---------------|--------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
|               | Cod e-Mix ed | Bang lish   | Engl ish    | Ban gla     | Cod e-Mix ed | Bang lish   | Engl ish    | Ban gla     |
| SVM           | 0.87         | 0.90        | 0.94        | 0.85        | 0.86         | 0.90        | 0.93        | 0.83        |
| Bi-LSTM       | 0.87         | 0.89        | 0.96        | 0.87        | 0.86         | 0.88        | 0.96        | 0.87        |
| mBERT         | 0.90         | 0.91        | 0.95        | <b>0.91</b> | 0.90         | 0.91        | 0.95        | 0.89        |
| XLM-RoBERT a  | <b>0.91</b>  | <b>0.91</b> | <b>0.96</b> | 0.90        | <b>0.91</b>  | <b>0.91</b> | <b>0.96</b> | 0.89        |
| Banglish BERT | 0.88         | 0.82        | 0.91        | 0.90        | 0.87         | 0.75        | 0.87        | <b>0.90</b> |
| DistilBE RT   | 0.90         | 0.91        | 0.96        | 0.91        | 0.89         | 0.91        | 0.96        | 0.90        |

TABLE VII. PERFORMANCE EVALUATION OF DIFFERENT MODELS FOR MULTILINGUAL EMOTION DETECTION

| Model        | Accuracy     |             |             |             | Weighted F1  |             |             |             |
|--------------|--------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
|              | Cod e-Mix ed | Bang lish   | Engl ish    | Ban gla     | Cod e-Mix ed | Bang lish   | Engl ish    | Ban gla     |
| SVM          | 0.64         | 0.66        | 0.65        | 0.61        | 0.56         | 0.62        | 0.61        | 0.55        |
| Bi-LSTM      | 0.65         | 0.64        | 0.64        | 0.64        | 0.61         | 0.57        | 0.61        | 0.60        |
| mBERT        | 0.67         | 0.65        | 0.66        | <b>0.67</b> | 0.66         | 0.64        | <b>0.65</b> | <b>0.65</b> |
| XLM-RoBERTa  | <b>0.68</b>  | <b>0.66</b> | <b>0.67</b> | 0.64        | <b>0.67</b>  | <b>0.65</b> | 0.63        | 0.62        |
| BanglishBERT | 0.66         | 0.59        | 0.57        | 0.65        | 0.62         | 0.43        | 0.41        | 0.64        |
| DistilBERT   | 0.67         | 0.65        | 0.65        | 0.66        | 0.65         | 0.64        | 0.64        | 0.64        |

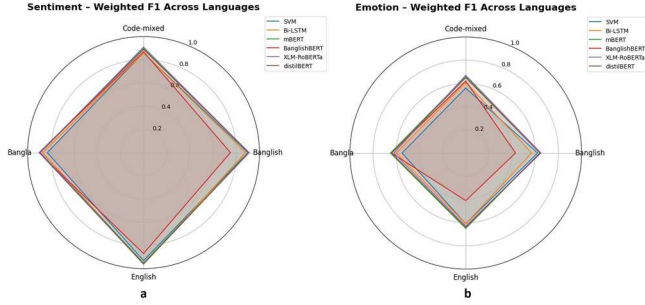


Fig. 6: Radar plot comparing model's performances using weighted F1 for different languages. (a) Sentiment Analysis (b) Emotion Detection

Fig. 6. depicts the radar plots based on weighted F1 score for all the models we trained across code-mixed, Banglish, English, & Bangla languages to observe the performance of the models in sentiment & emotion detection.

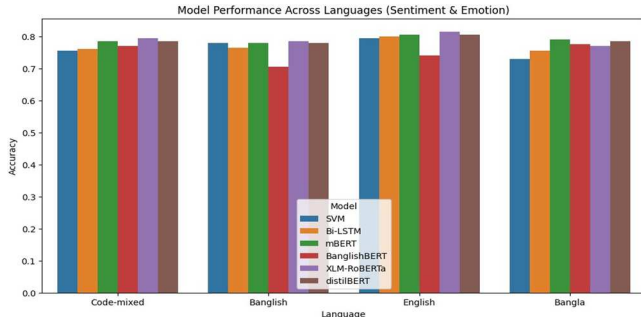


Fig. 7. Model performances across languages for both sentiment & emotion classification combined.

In Fig. 7, the bar chart showing the combined accuracy of the models across different languages. This is clearly noticeable in the bar chart that XLM-RoBERTa performed best for code-mixed, English & Banglish language where as mBERT & DistilBERT performed quite impressive for Bangla language.

#### D. Comparison of recent works

Table-VIII presents the comparative result between the proposed XLM-RoBERTa model and several available models for sentiment and emotion detection. Towards external validity, we also demonstrate a strict cross-lingual test with some other multilingual work to ensure broader robustness. The proposed model clearly outperforms the above-mentioned models while benchmarking strongly against previously done work. These combined reasons validate our proposed framework's usability and generalizability for real-life multilingual e-commerce scenarios.

TABLE VIII. PERFORMANCE COMPARISON OF XLM-RoBERTa WITH EXISTING MODELS

| Analysis  | Study                | Model                               | Task                                | Accura cy    | F1 score     |
|-----------|----------------------|-------------------------------------|-------------------------------------|--------------|--------------|
| Sentiment | Haque et al.[13]     | CLSTM                               | Multi-class Sentiment               | 85.8%        | 86%          |
|           | Alam et al.[5]       | CNN-Bi-LSTM using skip-BanglaBERT   | Bangla Sentiment                    | 90.24%       | 89%          |
|           | Tareq et al.[6]      | XGBoost with Fasttext               | Code-mixed sentiment                | ---          | 87%          |
|           | Manias et al.[11]    | XLM-R                               | Multilingual sentiment              | 75.85%       | 76.42%       |
|           | Rahman et al.[14]    | XLM-RSA                             | Multilingual reviews sentiment      | 91.9%        | 91.5%        |
|           | Our work             | XLM-RoBERTa                         | Multilingual Sentiment (Binary)     | <b>92.6%</b> | <b>92.4%</b> |
| Emotion   | Parvin & Hoque. [15] | Ensemble(T F + IDF)                 | Bangla emotion (multi-class)        | ----         | 62.39%       |
|           | Sadaf et al.[10]     | Logistic Regression (BOW + Unigram) | Bangla Emotion Detection            | 68.8%        | 65%          |
|           | Islam et al.[16]     | BanglaBERT                          | Bangla & Banglish Emotion detection | ---          | 62%          |
|           | Our work             | XLM-RoBERTa                         | Multilingual Emotion (5 class)      | <b>68.9%</b> | <b>69%</b>   |

#### E. Proposed Design & Framework

Our proposal is to build a smart real world e-commerce assistant which will combine both the sentiment & emotion understanding of the customer with smart features such as automatic response & customer engagement to help both the customer & seller. The system performs two key tasks: sentiment analysis (positive vs negative) and emotion classification across five categories (Happy, Love, Sadness, Fear, Anger). We evaluated six models across all languages for our task & XLM-RoBERTa outperformed the other models & can be chosen for further deployment. Our proposed architecture is illustrated in Fig. 8.

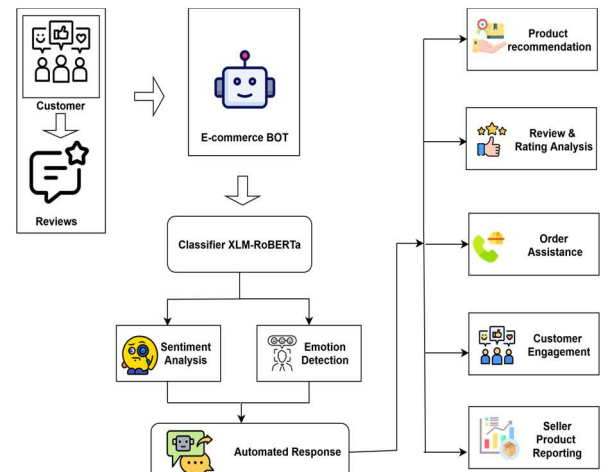


Fig. 8. Proposed E-commerce BOT framework based on sentiment and emotion analysis

When a customer submits a review then it goes to the sentiment and emotion modules. After evaluating the sentiment & emotion, the bot generates a context-aware reply and activates the other services. Here are the key features:

- Sentiment and Emotion Analysis: Detects positive or Negative sentiment and identifies emotional tone (Happy, Sadness, Love, Anger, Fear).
- Automated Response and Support Routing: Generates appropriate replies based on detected sentiment and emotion, and routes critical feedback to human agents.
- Product Recommendation and Engagement Engine: recommends item based on user preferences and emotional cues, supports loyalty offers, shows feedback prompts and personalize the follow-ups.
- Seller Product Reporting: Produces detailed analytical reports that summarizes product performance evaluating the sentiment & emotion metrics.
- Review and Rating Intelligence: Continuously monitors multilingual reviews across platforms to reveal trends in customer satisfaction.

The overall framework is cloud-driven, modular & designed for scalability and real time responsiveness, which makes it compatible to integrate with the e-commerce platforms.

#### F. Discussion & Analysis

After evaluating accuracy, precision, recall, and F1 score, XLM-RoBERTa achieved the highest performance, with 92.6% accuracy for sentiment classification and 68.9% for emotion detection. Although DistilBERT offered faster inference, its accuracy and F1 score were less stable, limiting suitability for real-time multilingual applications where consistent reliability is critical across diverse inputs. mBERT provided stronger results for Bangla but did not surpass XLM-RoBERTa in overall precision and F1. BanglishBERT also underperformed on Banglish subsets, reaching only 82% sentiment accuracy and 59% emotion accuracy. Considering these findings, XLM-RoBERTa is identified as the most suitable model for real-time deployment in a smart e-commerce assistant. Minor class imbalance in emotion detection remains a challenge, preventing performance from matching sentiment analysis.

#### IV. CONCLUSION

The research shows the performance of traditional machine learning & transformer-based models to detect sentiment & emotion of customer reviews to make a smart e-commerce assistant. XLM-RoBERTa is identified as the best model for both sentiment & emotion detection task across different languages. though mBERT & DistilBERT performs impressive but not reached XLM-RoBERTa as It has achieved the best accuracy among the other models which can be deployed & used to make a futuristic e-commerce smart Bot for real time analysis. Future research for balanced & larger dataset with linguistic diversity can be expanded for multilingual domain of ecommerce platforms to bring more efficiency.

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